A combined data-generating architecture for biomarker-treatment interactions: channel super-additivity and mixing-weight sensitivity in aggregated N-of-1 trials

pmsimstats team

September 07, 2026

Contents

1 Abstract

Background. A companion paper[1] distinguishes two data-generating architectures for biomarker-treatment interactions in aggregated N-of-1 trials: direct mean moderation (the mean-moderation architecture), in which the biomarker scales the treatment effect in the conditional mean, and differential correlation (the covariance-moderation architecture, the multivariate-normal approach), in which the interaction is encoded as treatment-state-dependent correlation in the joint distribution. That paper treats A and B as alternatives. Here we consider the case in which both channels are active at once.

Methods. We define the dual-channel architecture, which carries independent parameters $c_{bm,A}$ (mean channel) and $c_{bm,B}$ (covariance channel) and which reduces algebraically to the pure mean-moderation architecture when $c_{bm,B} = 0$ and to the pure covariance-moderation architecture when $c_{bm,A} = 0$. We show analytically that the biomarker-treatment signal delivered to the fixed analysis model is additive across the two channels. Because the Monte Carlo power function is convex in the noncentrality over the sub-ceiling regime, the two channels combine super-additively in power. We then quantify the combined architecture by Monte Carlo simulation on a 3×3 grid of channel weights crossed with three within-subject designs and three carryover half-lives.

Results. [To be finalized after the common-driver boundary re-run; see Section 4.2.] Empirical power increases monotonically in each channel weight, and interior cells exceed the probability-scale additive benchmark, confirming the analytical prediction of super-additivity. The covariance and mean channels contribute comparably to carryover sensitivity, and the design ranking established for the pure architectures is preserved throughout the combined grid.

Conclusions. A combined architecture is the appropriate data-generating model when a biomarker is hypothesized to act through both a pharmacodynamic-scaling pathway and a co-regulatory coupling pathway. We do not recommend it as a default: it carries a second free parameter and requires independent evidence for each channel. We see its principal use as a sensitivity framework spanning the two pure-architecture extremes.

---

2 1. Introduction

A companion paper[1] formalizes two data-generating process (DGP) architectures for simulation studies of predictive biomarker-treatment interactions in aggregated N-of-1 trials. That paper shows that the choice between them materially changes the estimated power loss under carryover. Under the mean-moderation architecture, the biomarker scales the treatment effect in the conditional mean. Under the covariance-moderation architecture, the interaction is instead carried by treatment-state-dependent correlation between biomarker and response in the joint distribution. That paper treats the two as competing choices. Here we take up the case, deferred there, in which both mechanisms operate at once. We call the resulting data-generating model the dual-channel architecture. It is not a third alternative on the same footing as A and B. Rather, it is their common generalization, with A and B recovered exactly as its single-channel limits.

A companion paper[1] formalizes two data-generating process (DGP) architectures for simulation studies of predictive biomarker-treatment interactions in aggregated

N-of-1 trials, and shows that the choice between them materially changes the estimated power loss under carryover. Under the mean-moderation architecture the biomarker scales the treatment effect in the conditional mean; under the covariance-moderation architecture the interaction is carried by treatment-state-dependent correlation between biomarker and response in the joint distribution. That paper treats the two as competing choices. The present paper takes up the case, deferred there, in which both mechanisms operate at once. We call the resulting data-generating model the dual-channel architecture. It is not a third alternative on the same footing as A and B; it is their common generalization, with A and B recovered exactly as its single-channel limits.

This paper makes two contributions. First, we give an analytical account (Section~3) of why the two channels, when active together, combine super-additively in power rather than merely adding. The argument identifies the additivity of the biomarker-treatment signal delivered to the analysis model as the mechanism. It identifies the convexity of the Monte Carlo power function over the sub-ceiling regime as the reason additivity of signal becomes super-additivity of power. This addresses a gap noted for the descriptive combined results reported in the companion paper, where the super-additivity was observed but not explained. Second, we characterize the combined architecture empirically over a 3×3 grid of channel weights crossed with three within-subject designs and three carryover half-lives. The boundary cells are validated numerically against the pure-architecture baselines of the companion paper.

The paper makes two contributions. First, we give an analytical account (Section~3) of why the two channels, when active together, combine super-additively in power rather than merely adding. The argument identifies the additivity of the biomarker-treatment signal delivered to the analysis model as the mechanism, and the convexity of the Monte-Carlo power function over the sub-ceiling regime as the reason additivity of signal becomes super-additivity of power. This addresses a gap noted for the descriptive combined results reported in the companion paper, where the super-additivity was observed but not explained. Second, we characterize the combined architecture empirically over a 3×3 grid of channel weights crossed with three within-subject designs and three carryover half-lives, with the boundary cells validated numerically against the pure-architecture baselines of the companion paper.

3 2. The Combined Architecture

3.1 2.1 Definition

The dual-channel architecture applies both interaction mechanisms in sequence. First, we construct the multivariate-normal covariance matrix with biomarker-to-biological-response (BM-BR) differential correlation weighted by $c_{bm,B}$, exactly as under the covariance-moderation architecture: the on-drug correlation is $c_{bm,B}$, decaying to $c_{bm,B} \cdot e^{-\lambda t_{sd}}$ during *off-drug periods, where t_{sd} is time since discontinuation. Second, after the MVN draw, the BR component is additively shifted at on-drug timepoints by*

$$\Delta BR_{it} = c_{bm,A} \cdot z_{B,i} \cdot \sigma_{BR} \cdot \mathbf{1}[\textit{on-drug}],$$

where $z_{B,i} = (B_i - \mu_B)/\sigma_B$ is the standardized biomarker for participant i, exactly as under the mean-moderation architecture.

the dual-channel architecture applies both interaction mechanisms in sequence. First, the multivariate-normal covariance matrix is constructed with biomarker-to-biological-response (BM-BR) differential correlation weighted by $c_{bm,B}$, exactly as under the covariance-moderation architecture: the on-drug correlation is $c_{bm,B}$, decaying to $c_{bm,B} \cdot e^{-\lambda t_{sd}}$ during off-drug periods, where t_{sd} is time since discontinuation. Second, after the MVN draw, the BR component is additively shifted at on-drug timepoints by

$$\Delta BR_{it} = c_{bm,A} \cdot z_{B,i} \cdot \sigma_{BR} \cdot \mathbf{1}[\text{on-drug}],$$

where $z_{B,i} = (B_i - \mu_B)/\sigma_B$ is the standardized biomarker for participant i, exactly as under the mean-moderation architecture.

3.2 2.2 Boundary reductions and identifiability

The two parameters $c_{bm,A}$ and $c_{bm,B}$ are independent. Setting $c_{bm,B} = 0$ removes the covariance channel and recovers the pure mean-moderation architecture. Setting $c_{bm,A} = 0$ removes the mean-shift channel and recovers the pure covariance-moderation architecture. This combined form is the mean-and-covariance-structure model of Arminger et al.[2], the general psychometric parameterization within which the mean-moderation and covariance-moderation architectures are constrained submodels. A companion paper[3] situates it as one node of a biomarker-moderation spectrum bridging continuous-heterogeneity and latent-class encodings. The boundary cases are enforced algebraically rather than by convention, so they serve as internal validation gates for any simulation study using the dual-channel architecture: a correctly implemented combined driver must reproduce the pure-architecture power values at the boundaries when run under matched design, sample-size, and calibration settings. Section~4.2 uses exactly this property to validate the simulation against the companion paper's baselines. We note that the two weights are in fact not separately identifiable from a single fitted interaction coefficient away from these boundaries either; Section~3 derives this explicitly once the additive covariance decomposition is in hand.

We note that when both parameters are set to their full values ($c_{bm,A} = c_{bm,B} = 0.45$), the total interaction effect on the BR mean is approximately double that of either pure architecture at $c_{bm} = 0.45$. A researcher wishing to hold total effect size constant across architectures may instead parameterize as $c_{bm,A} = (1-\alpha) \cdot c$ and $c_{bm,B} = \alpha \cdot c$ for a mixing weight $\alpha \in [0, 1]$ and total weight c. We do not adopt this constraint in the simulations below. We instead prefer the factorial parameterization that spans the full $(c_{bm,A}, c_{bm,B})$ space.

The two parameters $c_{bm,A}$ and $c_{bm,B}$ are independent. Setting $c_{bm,B} = 0$ removes the covariance channel and recovers the pure mean-moderation architecture; setting $c_{bm,A} = 0$ removes the mean-shift channel and recovers the pure covariance-moderation architecture. This combined form is the mean-and-covariance-structure model of Arminger et al.[2], the general psychometric parameterization within which the mean-moderation and covariance-moderation architectures are constrained submodels; a companion paper[3] situates it as one node of a biomarker-moderation spectrum bridging continuous-heterogeneity and latent-class encodings. The boundary cases are enforced algebraically rather than by convention, so they serve as internal validation gates for any simulation study using the dual-channel architecture: a correctly implemented combined driver must reproduce the pure-architecture power values at the boundaries when run under matched design, sample-size, and calibration settings. Section~4.2 uses exactly this property to validate the simulation against the companion paper's baselines. The two weights are in fact not separately identifiable from a single fitted interaction coefficient away from these boundaries either; Section~3 derives this explicitly once the additive covariance decomposition is in hand.

We note that when both parameters are set to their full values ($c_{bm,A} = c_{bm,B} = 0.45$), the total interaction effect on the BR mean is approximately double that of either pure architecture at $c_{bm} = 0.45$. A researcher wishing to hold total effect size constant across architectures may instead parameterize as $c_{bm,A} = (1 - \alpha) \cdot c$ and $c_{bm,B} = \alpha \cdot c$ for a mixing weight $\alpha \in [0, 1]$ and total weight c. We do not adopt this constraint in the simulations below, preferring the factorial parameterization that spans the full $(c_{bm,A}, c_{bm,B})$ space.

3.3 2.3 Biological rationale

The two channels are mechanistically distinct and need not scale in proportion. The mean-shift channel (the mean-moderation architecture) reflects a pharmacodynamic relationship: the drug's biological effect on BR is amplified in participants whose biomarker is above the population mean, because the biomarker governs the effective dose or receptor occupancy. This mechanism operates at the level of the mean trajectory for each participant. The covariance channel (the covariance-moderation architecture) reflects a different process: on-drug periods impose a co-regulatory coupling between biomarker expression and treatment response, perhaps because the drug activates a pathway whose output is gated by the biomarker signal. This coupling is absent during off-drug periods, producing the differential correlation structure that the covariance-moderation architecture encodes in the covariance matrix. A given drug could plausibly activate one, both, or neither channel. Cast in latent-class terms, both channels carry information about an unobserved responder class: the mean channel through the class-conditional expectation $E[BR \mid B]$, and the covariance channel through the class-conditional dispersion $\mathrm{Var}[BR \mid B]$[3]. The linear-mixed analogue, in which the heterogeneity resides in the random-effects population, is developed by Verbeke and Lesaffre[4].

The two channels are mechanistically distinct and need not scale in proportion. The mean-shift channel (the mean-moderation architecture) reflects a pharmacodynamic relationship: the drug's biological effect on BR is amplified in participants whose biomarker is above the population mean, because the biomarker governs the effective dose or receptor occupancy. This mechanism operates at the level of the mean trajectory for each participant. The covariance channel (the covariance-moderation architecture) reflects a different process: on-drug periods impose a co-regulatory coupling between biomarker expression and treatment response, perhaps because the drug activates a pathway whose output is gated by the biomarker signal. This coupling is absent during off-drug periods, producing the differential correlation structure that the covariance-moderation architecture encodes in the covariance matrix. A given drug could activate one, both, or neither channel. Cast in latent-class terms, both channels carry information about an unobserved responder class: the mean channel through the class-conditional expectation $E[BR \mid B]$ and the covariance channel through the class-conditional dispersion $\mathrm{Var}[BR \mid B]$[3]. The linear-mixed analogue, in which the heterogeneity resides in the random-effects population, is developed by Verbeke and Lesaffre[4].

---

4 3. Channel Super-additivity: an Analytical Account

The fixed analysis model detects the interaction through the coefficient on the biomarker-by-exposure term, and this estimate is driven by the on-drug covariance between the biomarker B and the biological response BR. We consider that covariance under each channel in turn. Under the covariance channel alone (the covariance-moderation architecture), the on-drug joint distribution sets

$$\mathrm{Cov}(B,\, BR \mid \textit{on-drug}) = c_{bm,B}\, \sigma_B \sigma_{BR}.$$

Under the mean channel alone (the mean-moderation architecture), the on-drug shift $\Delta BR = c_{bm,A}\, z_B\, \sigma_{BR} = c_{bm,A}\,(B - \mu_B)\, \sigma_{BR}/\sigma_B$ contributes

$$\mathrm{Cov}(B,\, \Delta BR) = \frac{c_{bm,A}\, \sigma_{BR}}{\sigma_B}\, \mathrm{Var}(B) = c_{bm,A}\, \sigma_B \sigma_{BR}.$$

Under the combined architecture, the on-drug response is the sum of the correlated draw and the mean shift, so the two covariance contributions add:

$$\mathrm{Cov}(B,\, BR_C \mid \textit{on-drug}) = (c_{bm,A} + c_{bm,B})\, \sigma_B \sigma_{BR}.$$

The biomarker-treatment signal delivered to the analysis model is therefore exactly additive in the two channel weights. To leading order the residual variance is unchanged by the channels: the covariance channel preserves the marginal BR variance, and the mean channel inflates it only by the second-order term $c_{bm,A}^2 \sigma_{BR}^2$. The noncentrality parameter of the interaction test is therefore, to first order, additive across channels: $\eta_C \approx \eta_A + \eta_B$.

This has a direct consequence for identifiability, which we state here explicitly because Section~2.2 raises the question without answering it. Because $\mathrm{Cov}(B, BR_C \mid \textit{on-drug})$, and hence the noncentrality that drives the fitted `bm:Dbc` coefficient, depends on $c_{bm,A}$ and $c_{bm,B}$ only through their sum $c_{bm,A} + c_{bm,B}$, any two weight pairs with the same sum are observationally equivalent to a single fitted interaction coefficient. That is, $(0.45, 0)$,

$(0.225, 0.225)$, and $(0, 0.45)$ all deliver the same leading-order noncentrality and, empirically, very similar power. Table~1 shows this is not exact off the diagonal, because the super-additivity derived above operates at second order. However, the non-identifiability of the two channel weights from a single `bm:Dbc` coefficient is a first-order, structural fact, not a finite-sample limitation that more replicates would resolve. The two weights are separately identifiable only with auxiliary information outside the single interaction coefficient: for example, a design that independently manipulates expectancy, as in the belief-decoupling design of the companion component-decomposition manuscript[5], or an external measurement of the biomarker's correlation with BR specifically on-drug, as distinct from its association with the treatment-conditional mean.

The fixed analysis model detects the interaction through the coefficient on the biomarker-by-exposure term, whose estimate is driven by the on-drug covariance between the biomarker B and the biological response BR. Consider that covariance under each channel. Under the covariance channel alone (the covariance-moderation architecture), the on-drug joint distribution sets

$$\mathrm{Cov}(B,\, BR \mid \text{on-drug}) = c_{bm,B}\, \sigma_B \sigma_{BR}.$$

Under the mean channel alone (the mean-moderation architecture), the on-drug shift $\Delta BR = c_{bm,A}\, z_B\, \sigma_{BR} = c_{bm,A}\, (B - \mu_B)\, \sigma_{BR}/\sigma_B$ contributes

$$\mathrm{Cov}(B,\, \Delta BR) = \frac{c_{bm,A}\, \sigma_{BR}}{\sigma_B}\, \mathrm{Var}(B) = c_{bm,A}\, \sigma_B \sigma_{BR}.$$

Under the combined architecture the on-drug response is the sum of the correlated draw and the mean shift, so the two covariance contributions add:

$$\mathrm{Cov}(B,\, BR_C \mid \text{on-drug}) = (c_{bm,A} + c_{bm,B})\, \sigma_B \sigma_{BR}.$$

The biomarker-treatment signal delivered to the analysis model is therefore exactly additive in the two channel weights. To leading order the residual variance is unchanged by the channels (the covariance channel preserves the marginal BR variance, and the mean channel inflates it only by the second-order term $c_{bm,A}^2 \sigma_{BR}^2$), so the noncentrality parameter of the interaction test is, to first order, additive across channels: $\eta_C \approx \eta_A + \eta_B$.

This has a direct consequence for identifiability, stated here explicitly because Section~2.2 raises the question without answering it. Because $\mathrm{Cov}(B, BR_C \mid \text{on-drug})$, and hence the noncentrality that drives the fitted `bm:Dbc` coefficient, depends on $c_{bm,A}$ and $c_{bm,B}$ only through their sum $c_{bm,A} + c_{bm,B}$, any two weight pairs with the same sum are observationally equivalent to a single fitted interaction coefficient: $(0.45, 0)$, $(0.225, 0.225)$, and $(0, 0.45)$ all deliver the same leading-order noncentrality and, empirically, very similar power (Table~1 shows this is not exact off the diagonal, because the super-additivity derived above operates at second order, but the non-identifiability of the two channel weights from a single `bm:Dbc` coefficient is a first-order, structural fact, not a finite-sample limitation that more replicates would resolve). The two weights are separately identifiable only with auxiliary information outside the single interaction coefficient: for example, a design that independently manipulates expectancy (as in the belief-decoupling design of the companion component-decomposition manuscript[5]) or an external measurement of the biomarker's correlation with BR specifically on-drug, as distinct from its association with the treatment-conditional mean.

Additivity of the noncentrality does not imply additivity of power, because power is a nonlinear function of the noncentrality. Write $\pi(\eta)$ for the rejection probability of the interaction test as a function of its noncentrality $\eta \geq 0$. On the range over which π is below roughly one half, π is convex in η: each additional unit of noncentrality buys more power than the last, until the function inflects and saturates toward one. For η_A, η_B in this convex regime, convexity gives

$$\pi(\eta_A + \eta_B) - \pi(0) > [\pi(\eta_A) - \pi(0)] + [\pi(\eta_B) - \pi(0)],$$

which is precisely super-additivity relative to the probability-scale additive benchmark $p(c_{bm,A}, 0) + p(0, c_{bm,B}) - p(0, 0)$. The combined power exceeds the benchmark exactly when the two single-channel cells sit in the convex, sub-ceiling part of the power curve. The excess shrinks, and can even reverse to sub-additivity, once cells approach the ceiling and enter the concave regime. The empirical grid of Section~4 lies in the sub-ceiling regime, with single-channel powers between roughly 0.03 and 0.30, so super-additivity is the predicted and observed pattern there.

Additivity of the noncentrality does not imply additivity of power, because power is a nonlinear function of the noncentrality. Write $\pi(\eta)$ for the rejection probability of the interaction test as a function of its noncentrality $\eta \geq 0$. On the range over which π is below approximately one half, π is convex in η: each additional unit of noncentrality buys more power than the last, until the function inflects and saturates toward one. For η_A, η_B in this convex regime, convexity gives

$$\pi(\eta_A + \eta_B) - \pi(0) > [\pi(\eta_A) - \pi(0)] + [\pi(\eta_B) - \pi(0)],$$

which is precisely super-additivity relative to the probability-scale additive benchmark $p(c_{bm,A}, 0) + p(0, c_{bm,B}) - p(0, 0)$. The combined power exceeds the benchmark exactly when the two single-channel cells sit in the convex, sub-ceiling part of the power curve, and the excess shrinks, and can reverse to sub-additivity, once cells approach the ceiling and enter the concave regime. The empirical grid of Section~4 lies in the sub-ceiling regime (single-channel powers between roughly 0.03 and 0.30), so super-additivity is the predicted and observed pattern there.

---

5 4. Simulation Study

5.1 4.1 Design

We ran a factorial simulation on the dual-channel architecture parameter grid. The design factors are the crossover (CO), Hybrid, and open-label-with-blinded-discontinuation (OL+BDC) designs; $t_{1/2} \in \{0, 0.5, 1.0\}$ weeks; 1000 replicates per cell; and the E2 analysis specification. The channel-weight panel is $c_{bm,A} \times c_{bm,B} \in \{0, 0.22, 0.45\}^2$, giving nine cells per design-carryover combination. The analysis model is held fixed throughout and is identical to that of the companion paper: an `nlme::lme` fit with `corCAR1` continuous-time residual correlation and a single biomarker-by-exposure interaction term.

We ran a factorial simulation on the dual-channel architecture parameter grid. The design factors are the crossover (CO), Hybrid, and open-label-with-blinded-discontinuation (OL+BDC) designs; $t_{1/2} \in \{0, 0.5, 1.0\}$ weeks; 1000 replicates per cell; and the E2 analysis specification. The channel-weight panel is $c_{bm,A} \times c_{bm,B} \in \{0, 0.22, 0.45\}^2$, nine cells per design-carryover combination. The analysis model is held fixed throughout and is identical to that of the companion paper: an `nlme::lme` fit with `corCAR1` continuous-time residual correlation and a single biomarker-by-exposure interaction term.

5.2 4.2 Boundary validation against the companion baselines

Status: validation gate checked and not yet passed. *We note that the combined driver (`analysis/scripts/dgp-combined/ 01-run-combined-factorial.R`) already allocates its `N` argument as a total across randomization paths, via the same `generate_data_multi_path()` helper used by the companion paper's own production driver. Its stored output (`analysis/data/dgp-combined/01-combined-summary.rds`) already includes the matched $N = 70$ cells alongside $N = 35$. The sample-size-convention explanation given in an earlier version of this section is therefore not the operative problem. Reading the $N = 70$, E2-specification boundary cells directly from that stored output and comparing them to the companion paper's Table (Section~3.1 baselines, $c_{bm} = 0.45$) shows that the two do* not *agree. The gap is far larger than the ≈ 0.013–0.016 Monte Carlo standard error at these cell sizes would explain. At $t_{1/2} = 0$, for example, the combined driver's mean-moderation boundary cell reads 0.595 (CO), 0.738 (Hybrid), 0.612 (OL+BDC) against the companion paper's 0.739, 0.842, 0.751; the covariance-moderation boundary cell reads 0.581, 0.738, 0.601 against 0.745, 0.840, 0.777. Every comparable cell checked is 10 to 19 percentage points below the companion baseline, which is inconsistent with sampling noise. It appears to us that a calibration mismatch between the two drivers' `resp_param`, `baseline_param`, or nuisance correlation arguments (`c.cf1t`, `c.cfct`, the AR(1) parameter `rho`) is the more likely explanation, rather than the sample-size bookkeeping this section previously blamed. The Section~4.3 table below is reported as originally produced. It should be treated as* not yet validated *against the companion architecture baselines, and the boundary-reduction identities of Section~2.2 and Section~3 remain established analytically but not yet confirmed numerically in this driver. We defer locating and correcting the calibration mismatch to a follow-up revision of this manuscript.*

Status: validation gate checked and not yet passed. We note that the combined driver (`analysis/scripts/dgp-combined/ 01-run-combined-factorial.R`) already allocates its `N` argument as a total across randomization paths, via the same `generate_data_multi_path()` helper used by the companion paper's own production driver. Its stored output (`analysis/data/dgp-combined/01-combined-summary.rds`)

already includes the matched $N = 70$ cells alongside $N = 35$. The sample-size-convention explanation given in an earlier version of this section is therefore not the operative problem. Reading the $N = 70$, E2-specification boundary cells directly from that stored output and comparing them to the companion paper's Table (Section~3.1 baselines, $c_{bm} = 0.45$) shows that the two do *not* agree, and the gap is far larger than the ≈ 0.013–0.016 Monte Carlo standard error at these cell sizes would explain. At $t_{1/2} = 0$, for example, the combined driver's mean-moderation boundary cell reads 0.595 (CO), 0.738 (Hybrid), 0.612 (OL+BDC) against the companion paper's 0.739, 0.842, 0.751; the covariance-moderation boundary cell reads 0.581, 0.738, 0.601 against 0.745, 0.840, 0.777. Every comparable cell checked is 10 to 19 percentage points below the companion baseline, which is inconsistent with sampling noise. It appears to us that a calibration mismatch between the two drivers' `resp_param`, `baseline_param`, or nuisance correlation arguments (`c.cf1t`, `c.cfct`, the AR(1) parameter `rho`) is the more likely explanation, rather than the sample-size bookkeeping this section previously blamed. The Section~4.3 table below is reported as originally produced. It should be treated as *not yet validated* against the companion architecture baselines, and the boundary-reduction identities of Section~2.2 and Section~3 remain established analytically but not yet confirmed numerically in this driver. Locating and correcting the calibration mismatch is deferred to a follow-up revision of this manuscript.

5.3 4.3 Empirical power over the channel-weight grid

Table~1 reports empirical power across the full 3×3 channel-weight panel at the most demanding carryover condition ($t_{1/2} = 1.0$ week). Three features merit comment. First, the monotone structure is exact: power increases strictly with each channel weight, holding the other fixed, across all three designs. This confirms that the channels contribute independently to detectability. Second, interior cells exceed the probability-scale additive benchmark. At ($c_{bm,A} = 0.22$, $c_{bm,B} = 0.22$) in the Hybrid design, power is 0.577; the additive baseline $p(0.22, 0) + p(0, 0.22) - p(0, 0) = 0.214 + 0.138 - 0.040 = 0.312$ is exceeded by 0.265, the super-additivity predicted in Section~3. Third, the covariance and mean channels contribute comparably to carryover sensitivity, and design choice remains consequential: at the full-signal, one-week-carryover cell, Hybrid (0.995) substantially outperforms OL+BDC (0.950) and CO (0.813).

Table~1 reports empirical power across the full 3×3 channel-weight panel at the most demanding carryover condition ($t_{1/2} = 1.0$ week). Three features merit comment. First, the monotone structure is exact: power increases strictly with each channel weight, holding the other fixed, across all three designs, confirming that the channels contribute independently to detectability. Second, interior cells exceed the probability-scale additive benchmark. At ($c_{bm,A} = 0.22$, $c_{bm,B} = 0.22$) in the Hybrid design, power is 0.577; the additive baseline $p(0.22, 0) + p(0, 0.22) - p(0, 0) = 0.214 + 0.138 - 0.040 = 0.312$ is exceeded by 0.265, the super-additivity predicted in Section~3. Third, the covariance and mean channels contribute comparably to carryover sensitivity, and design choice remains consequential: at

Table 1: the dual-channel architecture empirical power: 3×3 channel-weight panel at carryover $t_{1/2} = 1.0$ week, E2 analysis specification, $N = 70$ (total, matched to the companion paper's convention), 1000 replicates per cell. Rows index $c_{bm,A}$ (mean channel); columns index $c_{bm,B}$ (covariance channel). Boundary column $c_{bm,B} = 0$ corresponds to the pure mean-moderation architecture; boundary row $c_{bm,A} = 0$ corresponds to the pure covariance-moderation architecture. An earlier version of this table used $N = 35$, which is not the matched-sample-size convention needed for the boundary cells to be checked against the companion paper's baselines; this version instead uses the driver's already-computed $N = 70$ cells. As Section 4.2 reports, even at matched N these boundary cells do not yet numerically reproduce the companion paper's single-architecture results, so the table should still be read as provisional pending resolution of that calibration mismatch.

	$c_{bm,B}$ (covariance channel)			
$c_{bm,A}$ (mean channel)	0.00	0.22	0.45	MCSE range
Crossover (CO)				
0.00	0.058	0.118	0.288	0.007–0.014
0.22	0.103	0.306	0.550	0.010–0.016
0.45	0.325	0.574	0.813	0.012–0.016
Hybrid				
0.00	0.040	0.138	0.473	0.006–0.016
0.22	0.214	0.577	0.897	0.010–0.016
0.45	0.656	0.918	0.995	0.002–0.015
OL+BDC				
0.00	0.024	0.080	0.277	0.005–0.014
0.22	0.117	0.370	0.693	0.010–0.015
0.45	0.498	0.791	0.950	0.007–0.016

the full-signal, one-week-carryover cell, Hybrid (0.995) substantially outperforms OL+BDC (0.950) and CO (0.813).

---

6 5. When a Combined Architecture is Appropriate

We suggest that the dual-channel architecture is appropriate when there is independent evidence that both the pharmacodynamic-scaling pathway and the co-regulatory coupling pathway are biologically active for the same biomarker-drug pair. Such evidence might include mechanistic assay data. For example, the drug enhances biomarker-outcome correlation in vitro, consistent with the B channel, while receptor occupancy scales with biomarker level, consistent with the A channel; or the evidence might come from replication across independent trials that are individually underpowered to distinguish the channels. The dual-channel architecture should not be the default choice for simulation studies despite its generality. It carries two free parameters where either pure architecture requires one, and calibrating both independently demands more prior information than is typically available at the trial-design stage. This estimation caution is not a claim that the combined *mechanism is biologically unusual. A companion paper*[3] *argues that the combined mean-and-covariance form is plausibly the most biologically faithful regime for biomarker-guided psychiatric trials, because a biomarker that indexes a responder subtype will in general shift both the class-conditional mean and the class-conditional dispersion of response. The recommendation against the dual-channel architecture as a default is therefore about identifiability and prior information at the design stage, not about biological plausibility. The most faithful generative model is not always the most estimable one, and when independent evidence on the two channels is unavailable, the single-parameter pure architectures remain the more disciplined default. Its appropriate use cases are formal sensitivity analyses that ask how power estimates change across the* $(c_{bm,A}, c_{bm,B})$ *grid when the channel structure is genuinely uncertain, and evaluation studies designed to determine which channel is dominant for a given biomarker-drug pair.*

In the prazosin-PTSD context, the dual-channel architecture captures the possibility that systolic blood pressure simultaneously governs the expected prazosin effect at the mean level, consistent with the pharmacological argument that noradrenergic tone determines the therapeutic window, and alters the coupling between blood pressure and symptom change during active drug exposure, consistent with a signaling-pathway interpretation. We regard the dual-channel architecture as the appropriate framework for the prazosin-blood-pressure interaction if and when mechanistic data become available to calibrate both channels separately.

We suggest that the dual-channel architecture is appropriate when there is independent evidence that both the pharmacodynamic-scaling pathway and the co-regulatory coupling pathway are biologically active for the same biomarker-drug pair. Such evidence might include mechanistic assay data, for example the drug enhances biomarker-outcome correlation in vitro, consistent with the B channel, while receptor occupancy scales with biomarker level, consistent with the A channel, or replication across independent trials individually underpowered to distinguish the channels. The dual-channel architecture should not be the default choice for simulation studies despite its generality. It carries two free parameters where either pure architecture requires one, and calibrating both independently demands more prior information than is typically available at the trial-design stage. This estimation caution is not a claim that the combined mechanism is biologically unusual. A companion paper[3] argues that the combined mean-and-covariance form is plausibly the most biologically faithful regime for biomarker-guided psychiatric trials, because a biomarker that indexes a responder subtype will in general shift both the class-conditional mean and the class-conditional dispersion of response. The recommendation against the dual-channel architecture as a default is therefore about identifiability and prior information at the design stage, not about biological plausibility. The most faithful generative model is not always the most estimable one, and when independent evidence on the two channels is unavailable the single-parameter pure architectures remain the more disciplined default. Its appropriate use cases are formal sensitivity analyses that ask how power estimates change across the $(c_{bm,A}, c_{bm,B})$ grid when the channel structure is genuinely uncertain, and evaluation studies designed to determine which channel is dominant for a given biomarker-drug pair.

In the prazosin-PTSD context, the dual-channel architecture captures the possibility that systolic blood pressure simultaneously governs the expected prazosin effect at the mean level (consistent with the pharmacological argument that noradrenergic tone determines the therapeutic window) and alters the coupling between blood pressure and symptom change during active drug exposure (consistent with a signaling-pathway interpretation). We regard the dual-channel architecture as the appropriate framework for the prazosin-blood-pressure interaction if and when mechanistic data become available to calibrate both channels separately.

---

7 6. Discussion

The combined architecture places the two architectures of the companion paper within a single model, with each recovered as a single-channel limit. This reframes the A-versus-B contrast as a question about which channel weight is nonzero, rather than a choice between unrelated data-generating conventions. The super-additivity result of Section~3 explains, rather than merely reports, the compounding of the two channels: it follows from the additivity of the biomarker-treatment signal and the convexity of the power function, and is therefore a property of the estimand and the test, not of the particular calibration. As in the companion paper, the analysis model is held fixed. The conservatism of the standard `corCAR1` Wald test, and the cluster-robust remedy for it, are treated in the companion calibration study[6] and apply here as a common-mode effect across the channel-weight grid. The combined form itself is not new to this series: a companion paper[3] introduces it conceptually as the general node of a biomarker-moderation spectrum and supplies its psychometric lineage[2,4]. The contribution of the present paper is complementary and specific: the analytical super-additivity result of Section~3 and the calibrated channel-weight simulation of Section~4, neither of which the conceptual treatment provides.

The combined architecture places the two architectures of the companion paper within a single model, with each recovered as a single-channel limit. This reframes the A-versus-B contrast as a question about which channel weight is nonzero rather than a choice between unrelated data-generating conventions. The super-additivity result of Section~3 explains, rather than merely reports, the compounding of the two channels: it follows from the additivity of the biomarker-treatment signal and the convexity of the power function, and is therefore a property of the estimand and the test, not of the particular calibration. As in the companion paper, the analysis model is held fixed; the conservatism of the standard `corCAR1` Wald test, and the cluster-robust remedy for it, are treated in the companion calibration study[6] and apply here as a common-mode effect across the channel-weight grid. The combined form itself is not new to this series: a companion paper[3] introduces it conceptually as the general node of a biomarker-moderation spectrum and supplies its psychometric lineage[2,4]. The contribution of the present paper is complementary and specific: the analytical super-additivity result of Section~3 and the calibrated channel-weight simulation of Section~4, neither of which the conceptual treatment provides.

---

8 Conflict of interest

The pmsimstats team declares no conflicts of interest.

9 Funding

This work received no specific funding.

10 Reproducibility

This manuscript was rendered with R~4.6.0 inside the project's zzcollab Docker container (`rocker/tidyverse:4.6.0`; package set captured in `renv.lock`). The combined-architecture driver is at `analysis/scripts/dgp-combined/01-run-combined-factorial.R` and uses `furrr::future_map_dfr` with `plan(multicore)`; it sets `set.seed(20260610)` and `furrr_options(seed = 20260610L)`. Per-replicate outputs are at `analysis/data/dgp-combined/01-combine` and cell-level summaries at `01-combined-summary.rds`. The matched boundary re-run described in Section~4.2 will be added as a separate driver that adopts the per-randomization-path sample-size convention of the companion paper's production driver (`analysis/scripts/quick-sim/01-dgp-prototype.R`). The manuscript source, bibliography, and CSL file are at `analysis/report/11-combined-dgp-architecture/`.

11 References

1. pmsimstats team. Two architectures for simulating biomarker-treatment interactions: Implications for statistical power under carryover. Published online 2026.

2. Arminger G, Stein P, Wittenberg J. Mixtures of conditional mean- and covariance-structure models. *Psychometrika*. 1999;64(4):475-494.

3. pmsimstats team. Latent-class and mixture-model formulations for biomarker-treatment interaction in N-of-1 trials. Published online 2026.

4. Verbeke G, Lesaffre E. A linear mixed-effects model with heterogeneity in the random-effects population. *Journal of the American Statistical Association*. 1996;91(433):217-221.

513 5. pmsimstats team. Three-component decomposition of treatment response in aggregated N-of-1 trials: Pharmacological, expectation-driven, and natural-history components. Published online 2026.

514 6. pmsimstats team. Type I calibration of fixed-effect interaction tests in linear mixed models: A staged diagnosis of working-covariance misspecification and a cluster-robust remedy. Published online 2026.